NUTRIMO
AGRI SCIENCE

Crop Nutrition Guide

Feeding crops for yield, quality and soil health

The 4R principle

Effective nutrition follows four rules: the Right source, at the Right rate, at the Right time, in the Right place. Applying them reduces waste and raises the return on every bag of fertilizer.

Essential nutrients and their role

Nutrient	Main role	Common deficiency signs
Nitrogen (N)	Leaf and shoot growth, green colour	Pale yellowing of older leaves, stunted growth
Phosphorus (P)	Root development, flowering, energy transfer	Purplish leaves, poor root growth, delayed maturity
Potassium (K)	Water regulation, fruit quality, disease resistance	Scorching of leaf margins on older leaves
Calcium (Ca)	Cell wall strength, fruit firmness	Blossom-end rot, tip burn on young leaves
Magnesium (Mg)	Chlorophyll formation	Interveinal yellowing on older leaves
Sulphur (S)	Protein and oil formation	Uniform yellowing of young leaves
Zinc (Zn)	Enzyme activity, growth hormones	Small leaves, rosetting, khaira disease in paddy
Boron (B)	Pollination and fruit set	Flower drop, hollow or cracked fruits
Iron (Fe)	Chlorophyll synthesis	Interveinal yellowing of young leaves

Stage-wise feeding principles

- Establishment: Phosphorus-rich nutrition and bio-fertilizers for strong roots.
- Vegetative growth: Balanced NPK (e.g. 19:19:19) with adequate nitrogen.
- Flowering & fruit set: Phosphorus, potassium, calcium and boron (e.g. 00:52:34).
- Fruit development: High potassium (e.g. 13:00:45) and calcium for size, colour and shelf life.

Good practices

- Base decisions on a soil test, repeated every 2–3 years.
- Split nitrogen into multiple doses rather than one heavy application.
- Spray foliar nutrients in the early morning or late evening.
- Do not mix calcium fertilizers with phosphates or sulphates in the same tank.
- Add organic matter regularly to improve nutrient-use efficiency.

Doses are general guidance. Always read and follow the product label and leaflet, and consult a Nutrimo agronomist for crop- and region-specific recommendations.